

Solutions: Leah's scenario slide #69 (lecture 5):

- Suppose that Leah's utility function is given by $u(I) = I^{\frac{1}{2}}$, where I represents annual income in thousands of rands.
 - ① Is Leah risk loving, risk neutral, or risk averse? Explain.
 - ② Suppose that Leah is currently earning an income of R10,000 ($I = 10$) and can earn that income next year with certainty. She is offered an investment opportunity which will require her to leave her job, and it offers a .5 probability of earning R16,000, and a .5 probability of earning R5,000. Should she take the investment opportunity?
 - ③ In (2), would Leah be willing to buy insurance to protect against the variable income associated with the investment opportunity? If so, how much would she be willing to pay for that insurance? (Hint: What is the risk premium?)

Q1: We need to find the second derivative of her utility function:

First derivative = $u'(I) = \frac{d}{dI} (I^{1/2}) = \frac{1}{2} I^{-1/2}$

Second derivative = $u''(I) = \frac{d}{dI} \left(\frac{1}{2} I^{-1/2} \right) = \frac{1}{2} \cdot \left(-\frac{1}{2} I^{-3/2} \right) = -\frac{1}{4} I^{-3/2}$

Since second derivative <0, the utility function is concave, and Leah is risk averse.

Note: you can use the derivatives to prove this, or use the method in the slides (which may be easier).